

P3_Causal_Model_Training – Detailed Documentation

1. Project Overview

Project Title:

Causal Analysis of Cashback Incentives on Consumer Spending Behavior

Objective:

To determine whether **UPI cashbacks** cause people to **spend more or less per transaction**, and to uncover the **strategic purpose** behind this effect using causal inference techniques.

Core Research Question:

“Are UPI cashbacks the reason for people using UPI more these days?”

2. Background and Context

UPI platforms and digital wallets (e.g., PhonePe, Paytm, Google Pay) use **cashback offers** as marketing tools.

While traditional marketing assumes incentives boost spending, this project aims to **causally test** whether high cashback rates actually **increase or decrease** transaction amounts.

Using **causal inference** instead of correlation, we can determine **true cause-and-effect** relationships — answering *why* the spending changes, not just *how much* it changes.

3. Team Collaboration Summary

Component	Team Member Responsible
Data Cleaning & EDA	Teammates
Feature Engineering	Teammates
Causal DAGs & Initial Regressions	Teammates
Causal Modeling & Interpretation	You (Final Step)

Your mission was to **train the causal models**, derive **treatment effects**, and present **interpretable, strategic business insights**.

4. Dataset Summary

Datasets Used:

- `transactions_clean.csv`
- `user_level_data.csv`

Key Columns:

Variable	Description
<code>transaction_amount</code>	Value of the transaction (₹)
<code>total_incentive_rate</code>	Cashback or incentive rate (%)
<code>payment_type</code>	Payment method (UPI, card, etc.)

<code>product_category</code>	Type of transaction (bill, recharge, etc.)
<code>transaction_hour</code>	Time of transaction
<code>location,</code> <code>device_type</code>	Contextual confounders

Data Size: ~5,000 transactions (approx.)

5. Causal Framework

Treatment Variable

`high_incentive` = 1 if `total_incentive_rate` > median incentive rate
(Defines whether a transaction received a “high cashback” offer.)

Outcome Variable

`transaction_amount` = Total rupee amount spent in that transaction

Confounders

`product_category`, `payment_type`, `location`, `device_type`,
`transaction_hour`, `high_value_txn`

6. Methodology

Step 1 – Propensity Score Estimation

Model used: **Logistic Regression**

Purpose:

To estimate the probability (propensity score) that a transaction receives a high cashback, based on confounding variables.

Output:

Each transaction receives a **propensity score (0–1)** representing its likelihood of being treated (high incentive).

Model accuracy: **70.0%**

Step 2 – Causal Estimation via Propensity Score Matching

Tool: `causalinference` library

Estimator: **Nearest-neighbor matching with bias adjustment**

Purpose:

To isolate the **Average Treatment Effect (ATE)** — the difference in transaction amount *caused* by receiving a high cashback.

Formula:

$$\begin{bmatrix} \\ \text{ATE} = E[Y(1) - Y(0)] \\ \end{bmatrix}$$

where

$Y(1)$ = outcome if treated (high cashback)

$Y(0)$ = outcome if untreated (low cashback)

Step 3 – Subgroup Analysis by Payment Method

Separate causal models were trained per payment method:

- Debit Card
- UPI

- Wallet Balance
- Credit Card
- Bank Transfer

Each subgroup model computed:

- Average Treatment Effect (ATE)
- Standard Error (SE)
- p-value (significance)
- Confidence Interval (95%)

7. Results

7.1 Overall Causal Result

Metric	Value
ATE	₹-1,867.14
95% CI	(₹-2,057 to ₹-1,677)
Significance	p < 0.001
Interpretation	High cashback causes lower spending per transaction

7.2 Breakdown by Payment Method

Payment Method	ATE (₹)	p-value	Interpretation	Rank
Debit Card	-1,575.82	5.7e-11	Decrease	1
UPI	-1,692.33	< 0.001	Decrease	2
Wallet Balance	-1,748.38	9.4e-14	Decrease	3
Credit Card	-1,874.94	6.6e-16	Decrease	4
Bank Transfer	-1,899.17	0.000	Decrease	5

All results statistically significant (p < 0.001).

7.3 Model Quality

- Propensity Score Accuracy: **70.0%**
- Robustness: Consistent across all payment methods
- All ATEs significant at 95% confidence level

8. Causal Interpretation

Key Finding

Cashbacks cause **lower** transaction amounts — but this is **by design**.

Platforms intentionally use **high cashbacks** to encourage **small, first-time** transactions for **user acquisition**, not to increase per-user spending.

The Mechanism



Low Large, Routine → Profit Retention
Cashback Transactions

This pattern gives the **illusion** that “high cashback = low spending,” but in reality, it’s a **smart marketing strategy** — a **strategic loss leader**.

9. Visualization Highlights

1. **Bar Chart:** Negative causal effect across all payment methods
2. **Comparison Chart:** High vs. low incentive transaction amounts
3. **Pie Chart:** Cashback strategy distribution (85% user acquisition focus)
4. **UPI Comparison Chart:** UPI vs. overall causal effect

(Visuals demonstrate that cashback decreases spending but boosts user acquisition.)

10. Business Recommendations

For Platforms (PhonePe, Paytm, Google Pay, etc.):

- Continue cashback-based **user acquisition** strategy
- Don’t rely on cashback for increasing spending volume
- Target high cashbacks to **new users or small transactions**

For Partner Businesses:

- Design campaigns focused on **acquiring new customers**, not upselling
- Measure **user growth KPIs**, not just transaction value

For Consumers:

- Best cashback deals occur on **small or first-time** transactions
 - Expect lower rewards on **large purchases**
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11. Executive Summary

UPI cashbacks are proven to cause lower transaction values because they are strategically used as loss leaders to acquire new users, not to increase spending from existing ones.

- **Causal Effect (UPI):** ₹-1,692 per transaction
 - **Significance:** $p < 0.001$
 - **Model Accuracy:** 70%
 - **Conclusion:** Cashbacks drive adoption, not revenue growth
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12. Final Deliverables

Output	File Name
Enhanced causal results	P3_Enhanced_Causal_Results.csv

Executive summary report	<code>P3_Executive_Summary.txt</code>
Visualization dashboard	Matplotlib-based multi-chart figure
Trained causal model code	<code>P3_Causal_Model_Training.ipynb</code>

13. Final Conclusion

Definitive Causal Answer:

High cashback rates do not increase user spending — they successfully **acquire** users.

Strategic Insight:

Cashbacks are **Strategic Loss Leaders** — short-term losses for long-term user growth.

Business Impact:

Platforms are already optimizing this perfectly; no change in strategy needed.

14. References

- Causal Inference in Python (`causal inference`, `causalml`)
- Rubin, D.B. (1974). *Estimating causal effects of treatments in randomized and nonrandomized studies*.
- Hernán & Robins (2020). *Causal Inference: What If*.

- Dataset and cleaning provided by team.

